



IT 402 — APPLIED FORECASTING METHODS

Rossmann Store Sales Forecasting

Group 15

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Problem Statement

Why accurate sales forecasting matters for retail chains

Retail stores face challenges predicting daily sales due to promotions, holidays, competition and seasonality.

Project Objectives

- Build forecasting models to predict daily sales for all Rossmann stores and for a particular store also
- Perform uncertainty analysis by predicting sales ranges (quantile regression)

Business Use Cases



Inventory Management

- Maintain optimal stock levels
- Reduce wastage and stockouts



Workforce Planning

- Schedule staff efficiently based on expected demand



Promotion Strategy

- Evaluate promotion effectiveness
- Plan future campaigns



Revenue Optimization

- Identify peak sales periods
- Maximize profit through planning



Store Level Decisions

- Compare store performance
- Find underperforming locations



Dataset Overview

Rossmann Store
Sales Dataset

1115 stores

Daily records of all
stores from Jan 2013
to June 2015

KEY FEATURES

Store

Unique store identifier

Date

Date of sales record

Sales

Daily revenue (target variable)

Customers

Number of customers visiting

Open

Whether the store was
open/closed

Promo

Active promotion flag

StateHoliday

Public holiday indicator

SchoolHoliday

School holiday indicator



Time Series Models

STATIONARITY

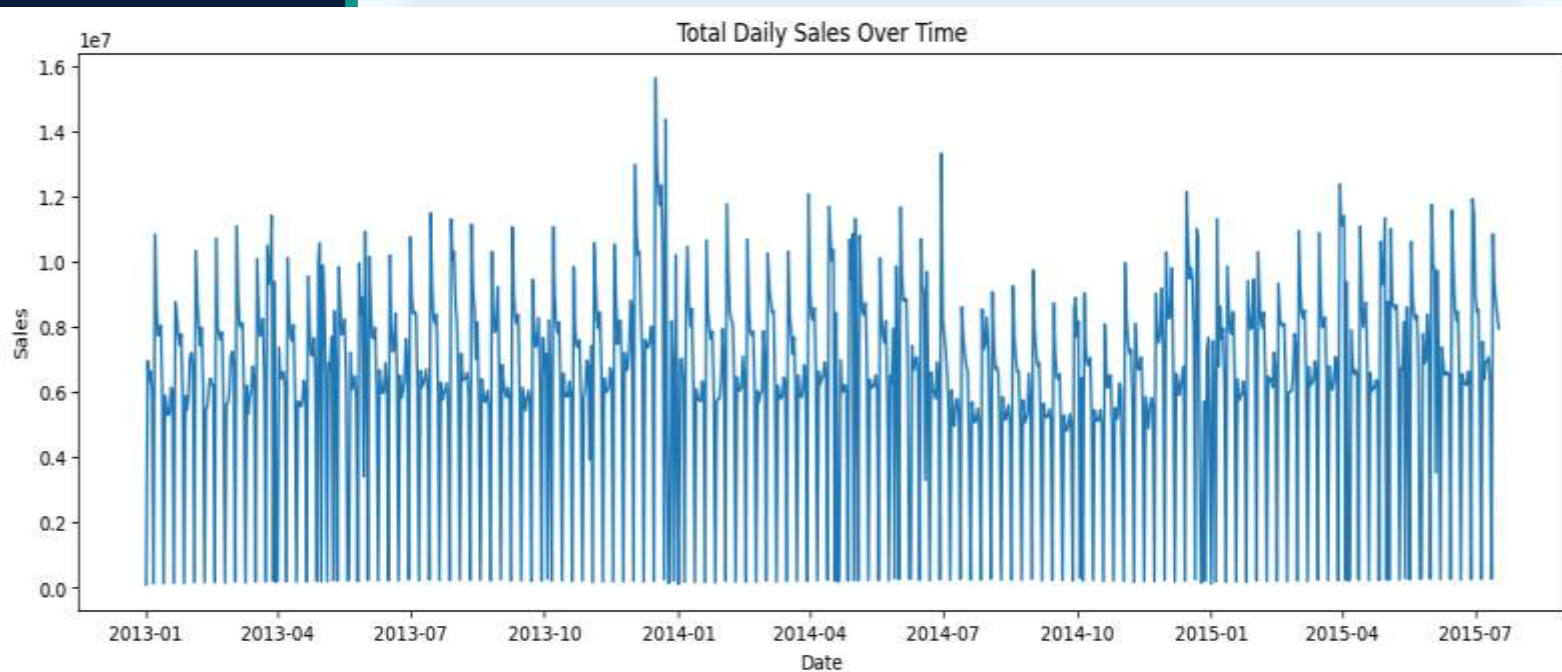
ADF Test

ADF Statistic: -4.58

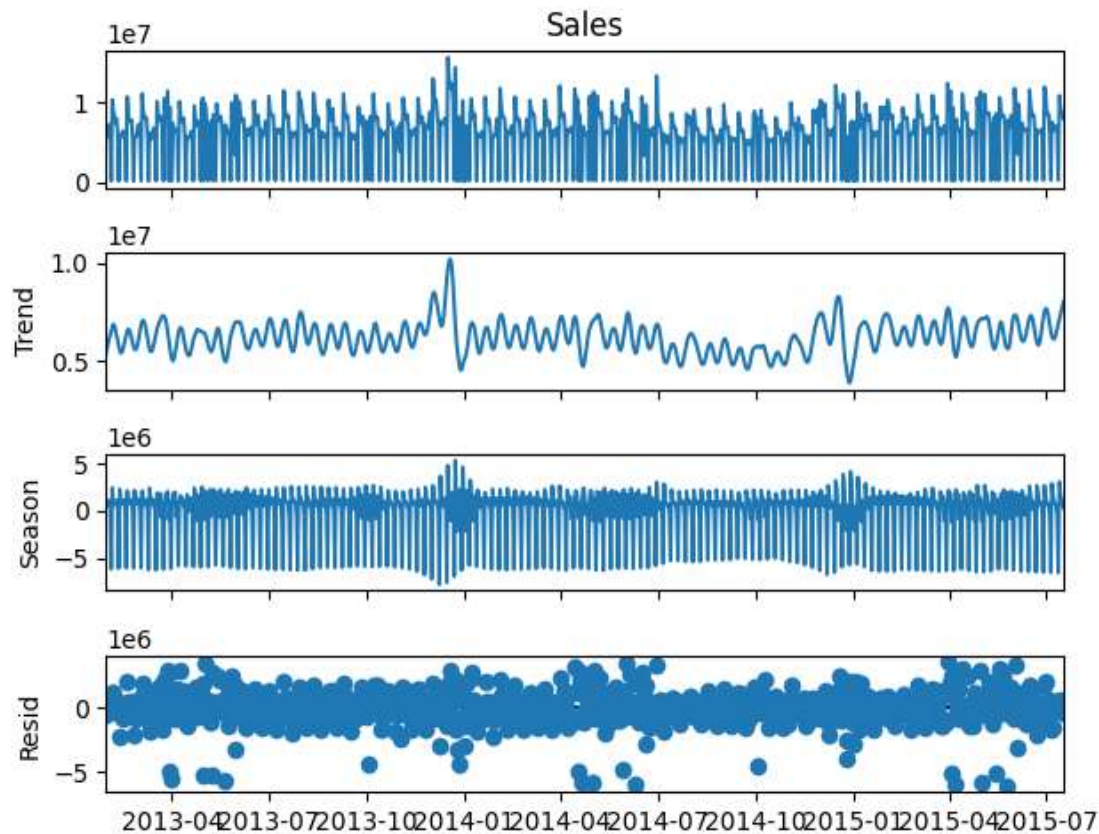
p-value: 0.000139

So the series is stationary.

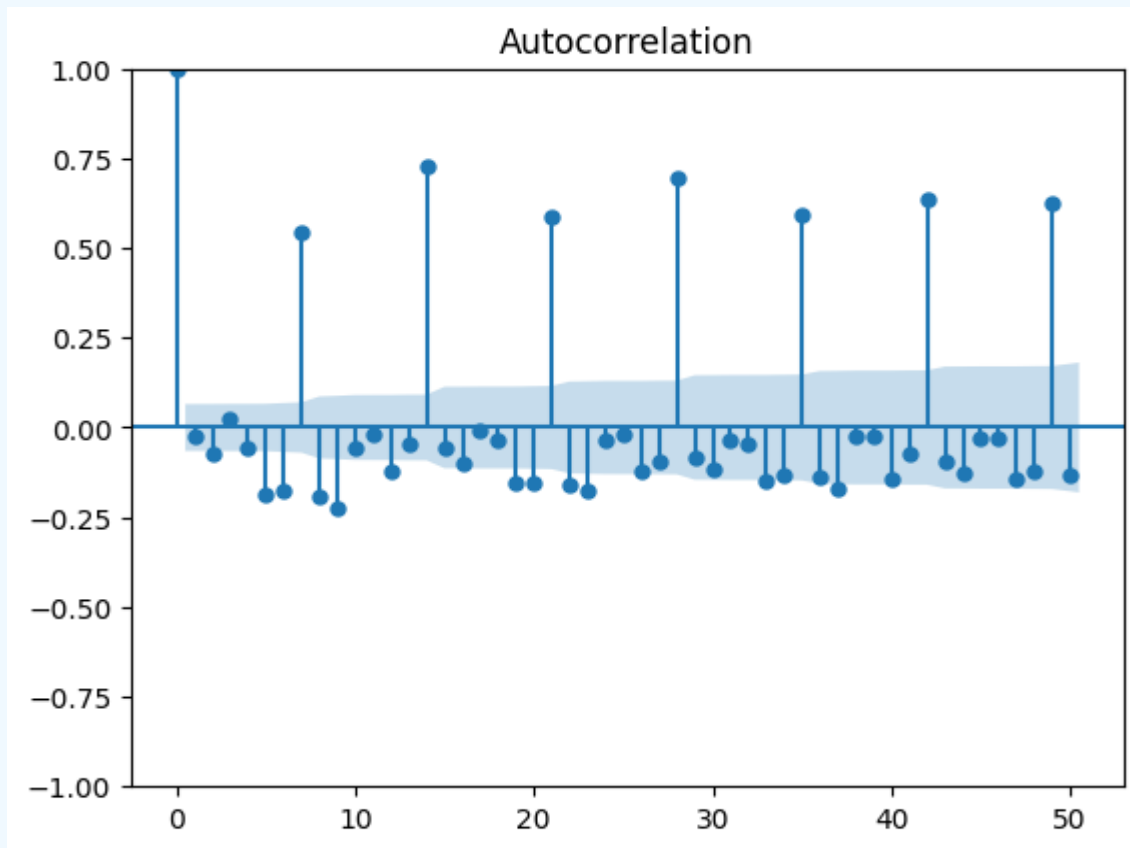
Stationarity analysis



STL Decomposition



ACF Analysis



Model Input, Output & Feature Engineering

INPUT FEATURES

Temporal:

Day of Week (0 - 6)
Month (1 - 12)
Year
Cyclical sin/cos encodings

Lag Features:

lag_1 (prev day sales)
lag_7 (1 week ago)
lag_14 (2 weeks ago)

Sequence Structure (Deep Learning):

Window = 60 days of
history → Predict Day
61

Exogenous:

Promo flag
SchoolHoliday flag
CompetitionDistance
IsWeekend flag



OUTPUT

Point Forecast:

Predicted daily sales on a given day

Prediction Intervals (Quantile):

q = 0.025	Lower bound (2.5th percentile)
q = 0.500	Median — central estimate
q = 0.975	Upper bound (97.5th percentile)



SARIMAX Model

Seasonal ARIMA
with exogenous
variables

CLASSICAL STATISTICAL MODEL

Model Parameters $(p, d, q) \times (P, D, Q, s)$

Parameter	Value	Description
p / P	[1 , 1]	AR / Seasonal AR order
d / D	[0 , 0]	Differencing / Seasonal diff
q / Q	[1 , 1]	MA / Seasonal MA order
s	[7]	Seasonal period

Evaluation Metrics

RMSE: 1.62

MAE: 1.15

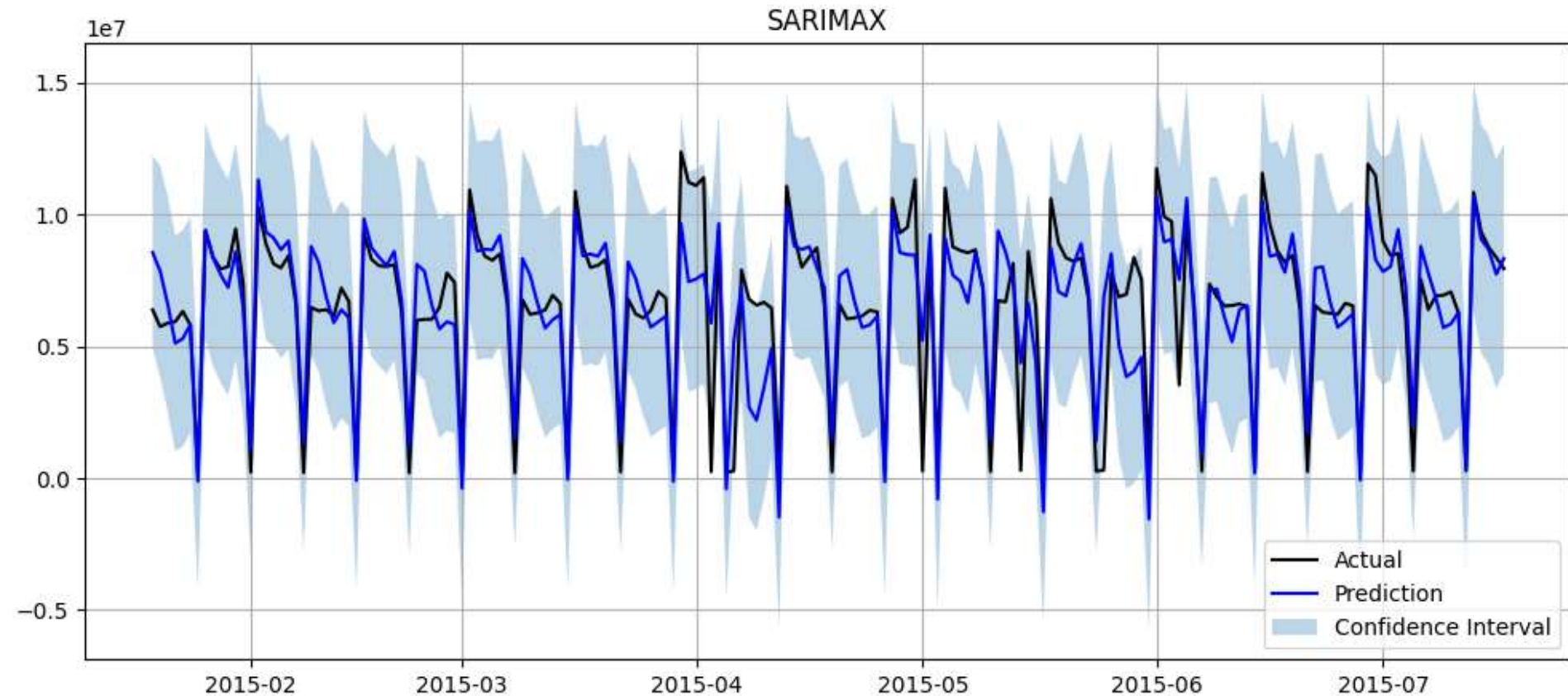
R2: 0.74

PICP: 96.66%

MPIW : 0.84

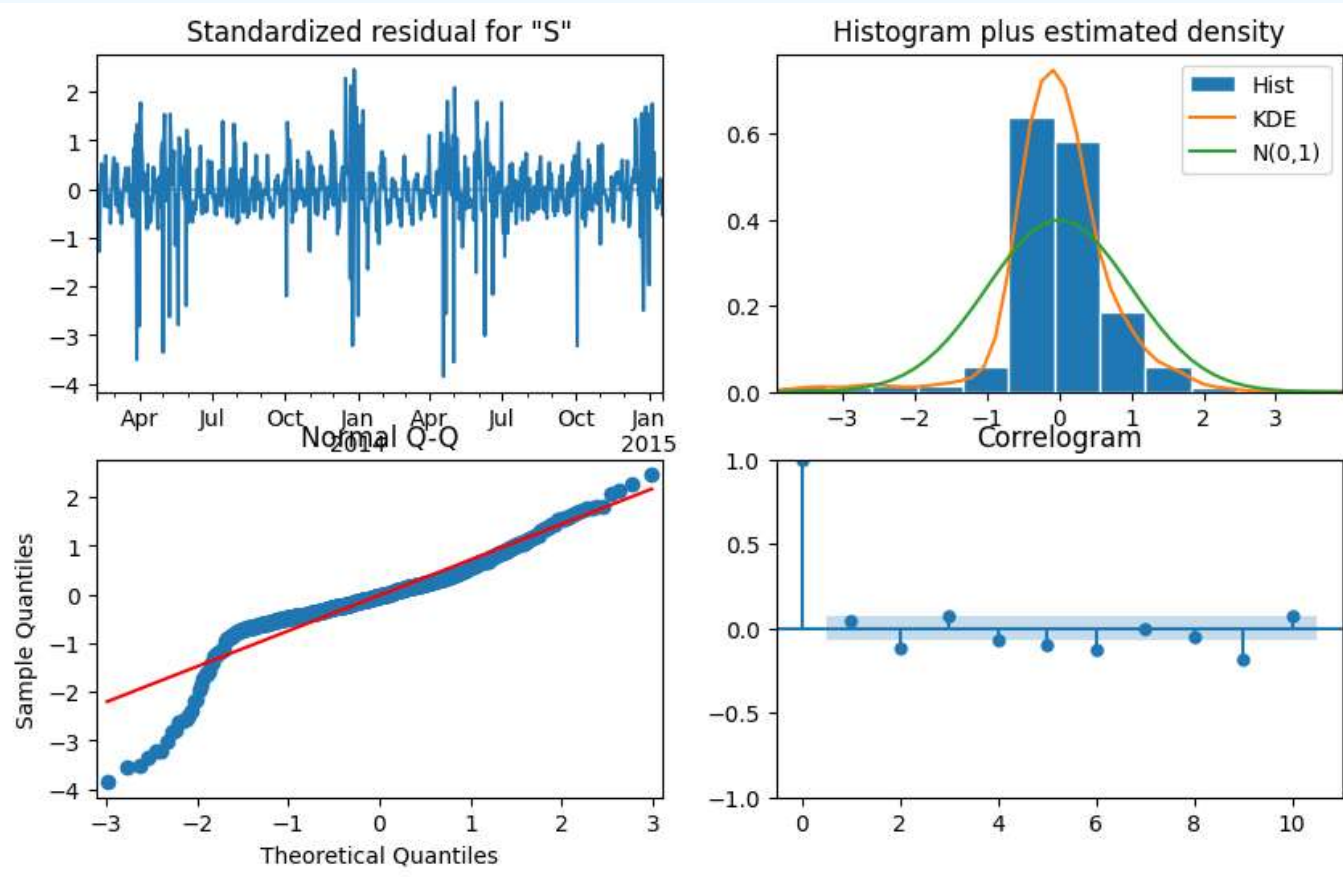


SARIMAX Model Prediction





SARIMAX Model Residual Analysis





Prophet Model

Evaluation Metrics

RMSE: 1.61

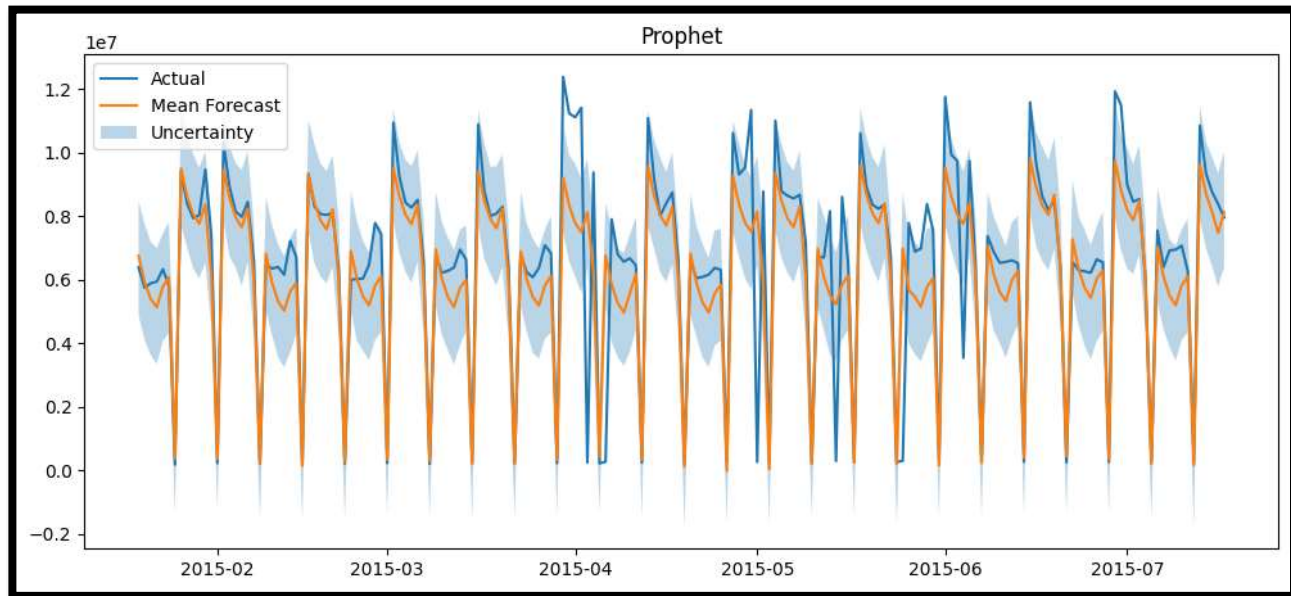
R2: 0.74

MAE: 0.96

PICP: 87.22%

MPIW: 0.4

Meta's open source forecasting model





Deep Learning Models - FNN (MLP)

Model: "FNN_Model"

Layer (type)	Output Shape	Param #
dense_71 (Dense)	(None, 128)	169,472
dropout_57 (Dropout)	(None, 128)	0
dense_72 (Dense)	(None, 64)	8,256
dropout_58 (Dropout)	(None, 64)	0
dense_73 (Dense)	(None, 32)	2,080
dense_74 (Dense)	(None, 1)	33

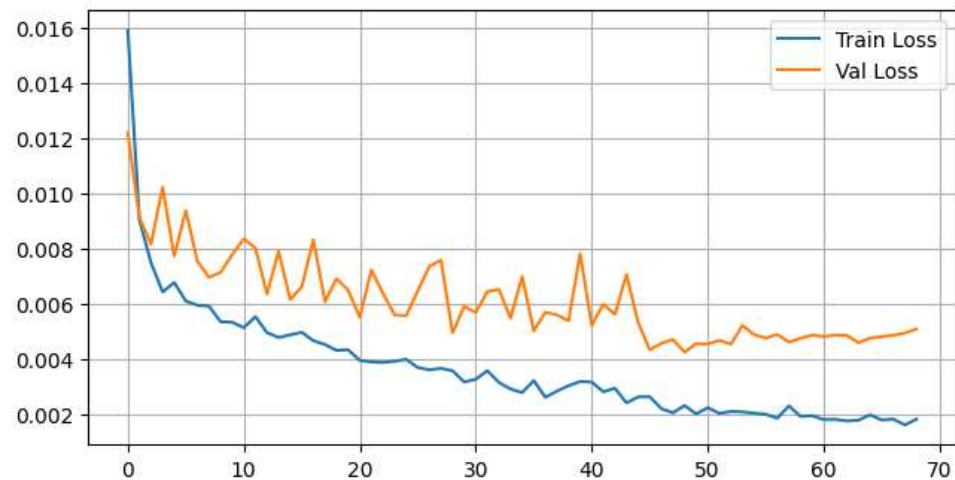
Evaluation Metrics : RMSE: 1.4 | MAE: 0.89 | R2: 0.81



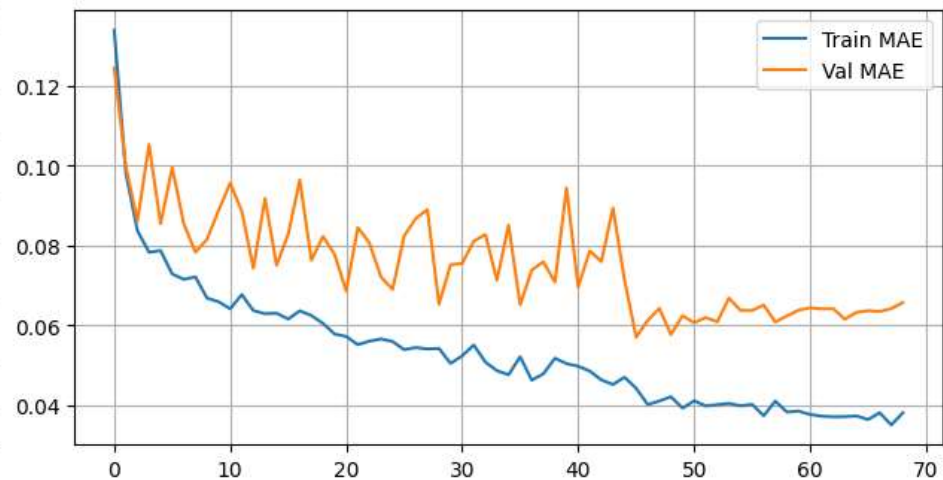
FNN Loss

FNN Training History

FNN — Loss

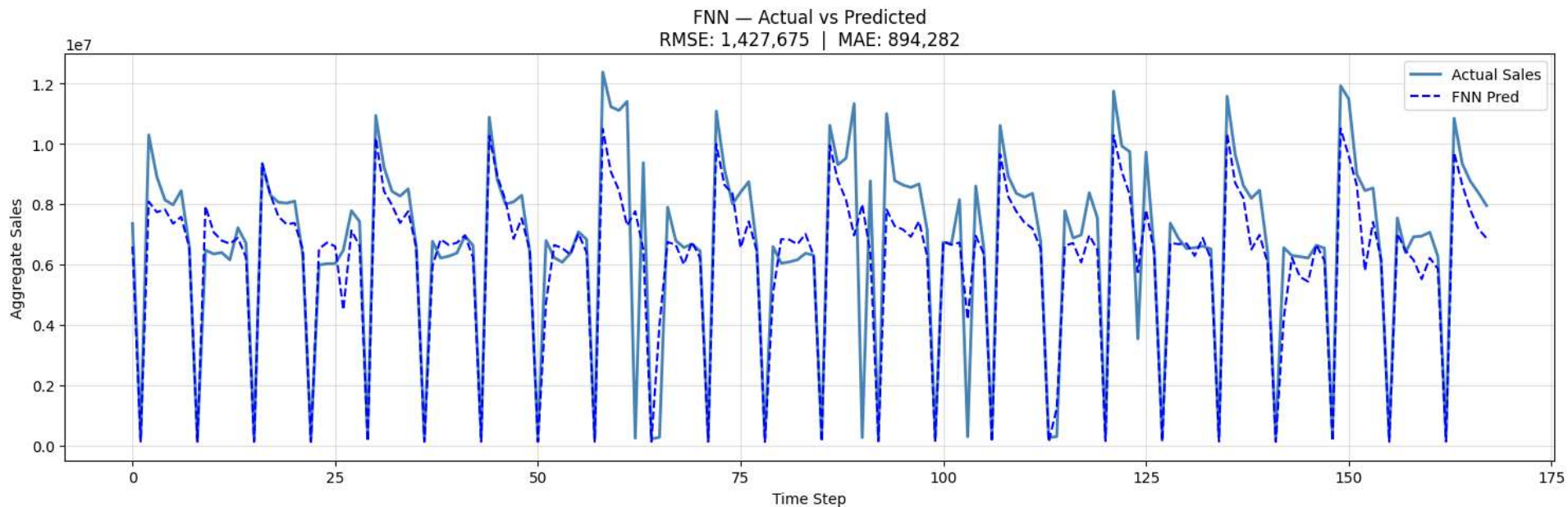


FNN — MAE





FNN Prediction





LSTM

Evaluation Metrics

- RMSE: 1.51
- MAE: 0.81
- R2: 0.79

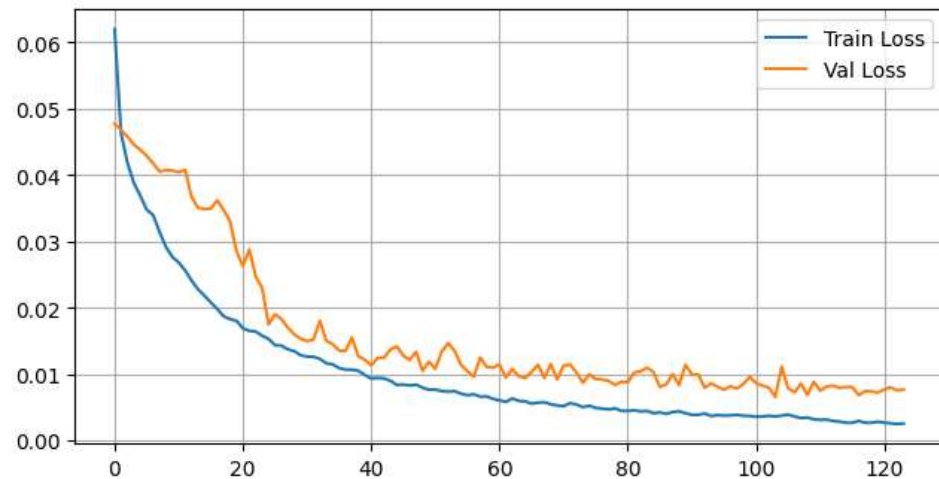
Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 60, 22)	0	-
lstm_4 (LSTM)	(None, 60, 128)	77,312	seq_input[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 60, 128)	512	lstm_4[0][0]
dropout_61 (Dropout)	(None, 60, 128)	0	batch_normalizat...
lstm_5 (LSTM)	(None, 64)	49,408	dropout_61[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 64)	256	lstm_5[0][0]
dropout_62 (Dropout)	(None, 64)	0	batch_normalizat...
dense_77 (Dense)	(None, 32)	2,080	dropout_62[0][0]
future_input (InputLayer)	(None, 3)	0	-
concatenate_12 (Concatenate)	(None, 35)	0	dense_77[0][0], future_input[0][...
dense_78 (Dense)	(None, 1)	36	concatenate_12[0...



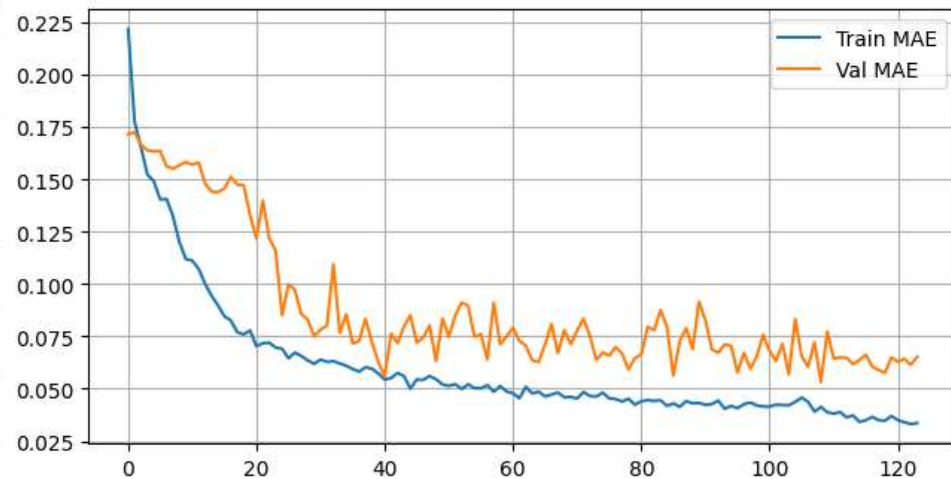
LSTM Loss

LSTM Training History

LSTM — Loss

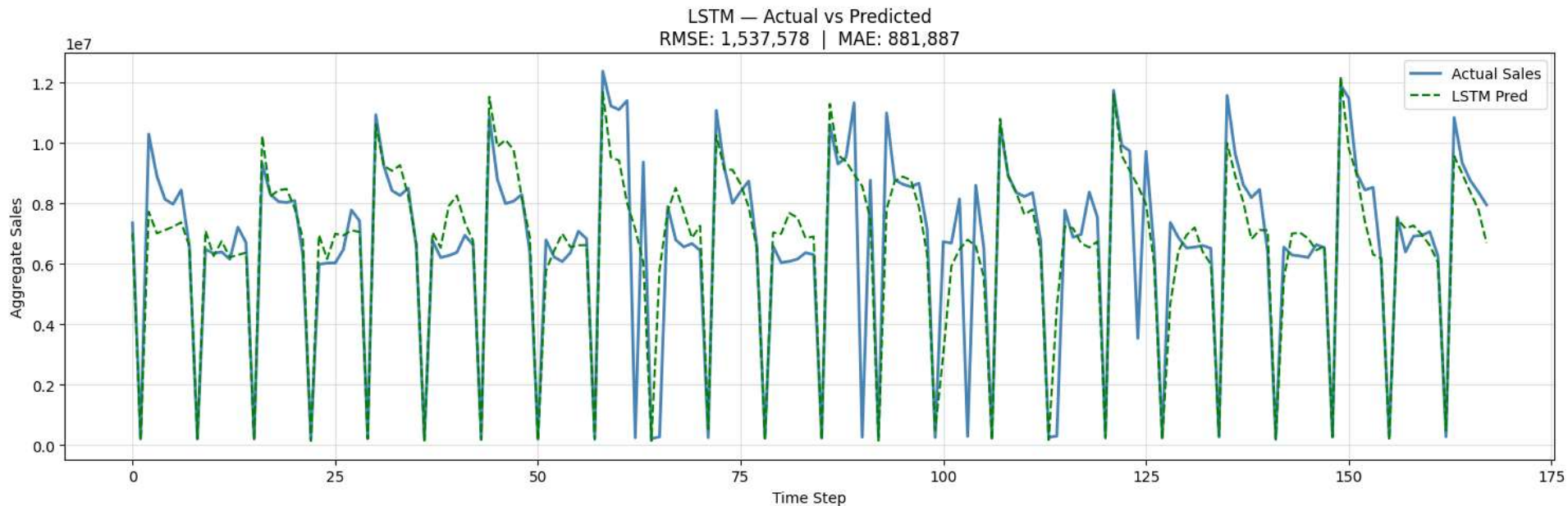


LSTM — MAE





LSTM Prediction





RNN

Evaluation Metrics

- RMSE: 1.54
- MAE: 0.87
- R2: 0.78

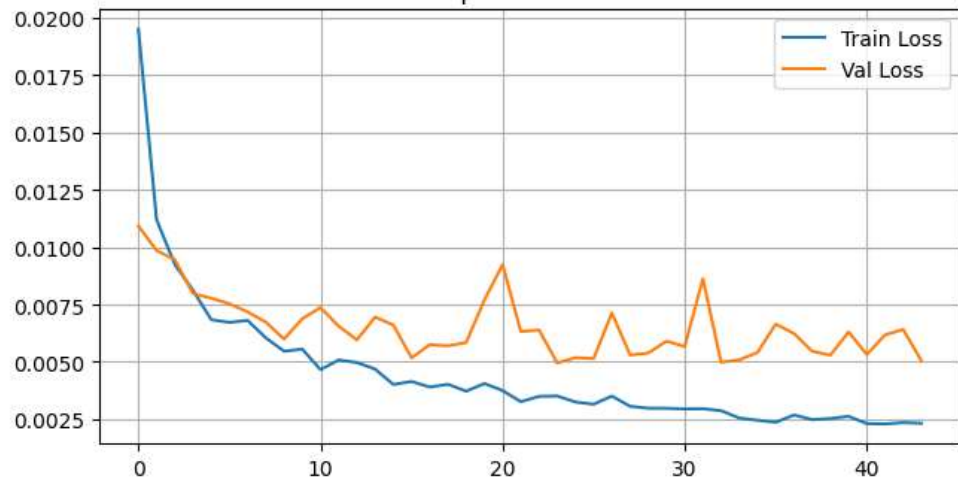
Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 60, 22)	0	-
simple_rnn_4 (SimpleRNN)	(None, 60, 128)	19,328	seq_input[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 60, 128)	512	simple_rnn_4[0][...]
dropout_59 (Dropout)	(None, 60, 128)	0	batch_normalizat...
simple_rnn_5 (SimpleRNN)	(None, 64)	12,352	dropout_59[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 64)	256	simple_rnn_5[0][...]
dropout_60 (Dropout)	(None, 64)	0	batch_normalizat...
dense_75 (Dense)	(None, 32)	2,080	dropout_60[0][0]
future_input (InputLayer)	(None, 3)	0	-
concatenate_11 (Concatenate)	(None, 35)	0	dense_75[0][0], future_input[0][...]
dense_76 (Dense)	(None, 1)	36	concatenate_11[0...



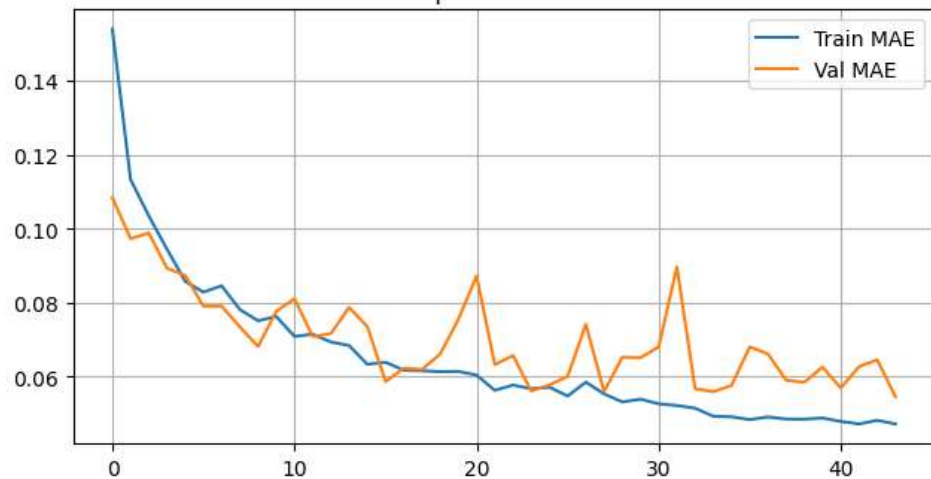
RNN Loss

SimpleRNN Training History

SimpleRNN — Loss

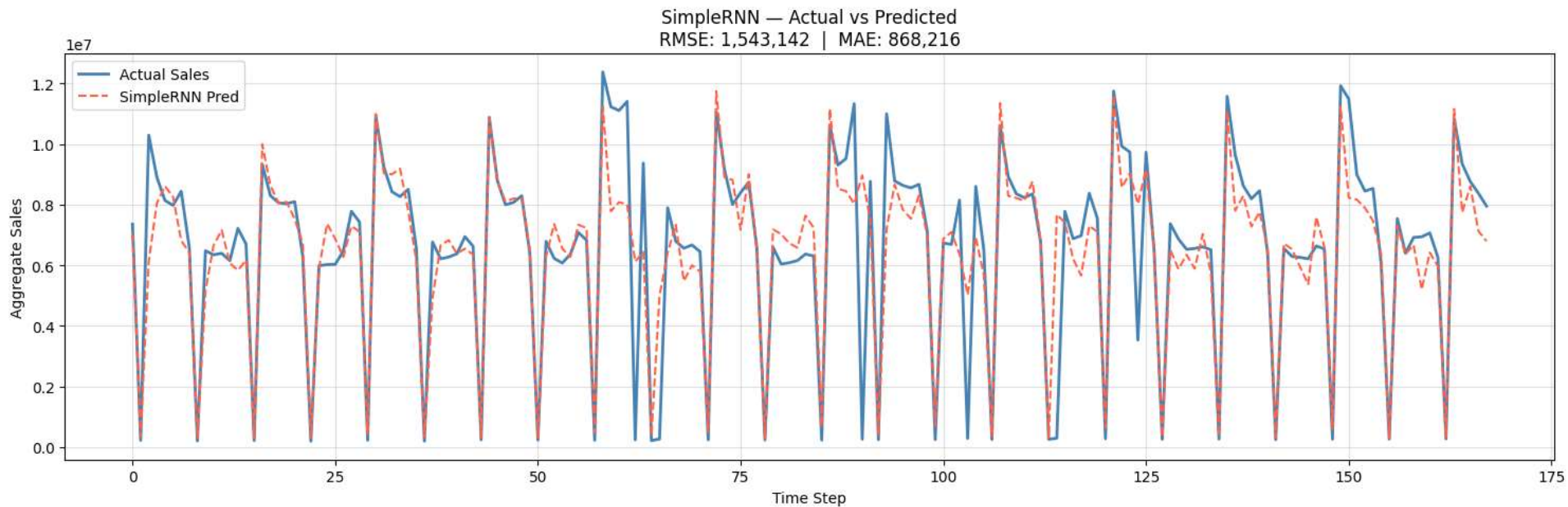


SimpleRNN — MAE





RNN Prediction





N-HiTS (Neural Hierarchical Interpolation for Time Series)

- It is a deep learning model specifically designed for time series forecasting.
- It predicts future values by breaking the time series into multiple parts and learning patterns at each level.
- It is fast like MLP.
- It captures long term patterns like LSTM.
- It handles complex seasonality better.

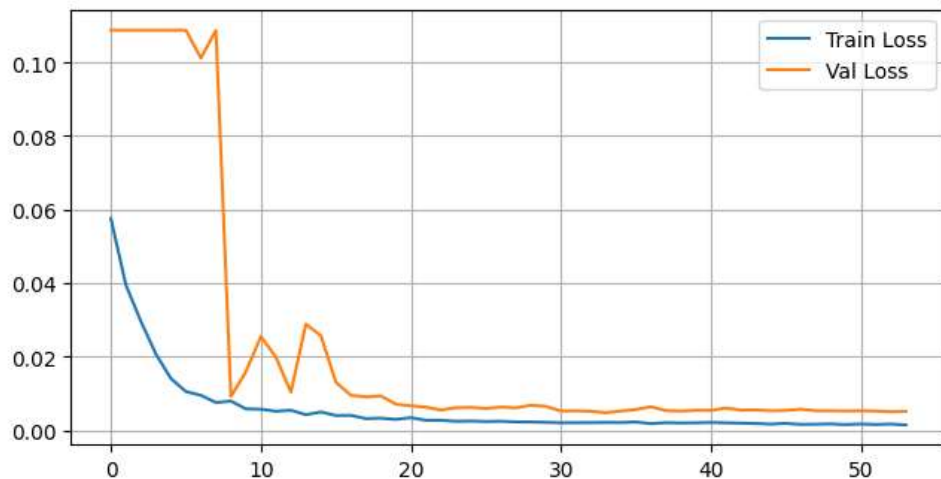
Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 60, 22)	0	-
flatten_44 (Flatten)	(None, 1320)	0	seq_input[0][0]
dense_93 (Dense)	(None, 1320)	1,743,720	flatten_44[0][0]
reshape_6 (Reshape)	(None, 60, 22)	0	dense_93[0][0]
n_hi_ts_block_6 (NHiTSBlock)	[(None, 1320), (None, 64)]	761,784	reshape_6[0][0]
subtract_6 (Subtract)	(None, 1320)	0	flatten_44[0][0], n_hi_ts_block_6[0][0]
dense_98 (Dense)	(None, 1320)	1,743,720	subtract_6[0][0]
reshape_7 (Reshape)	(None, 60, 22)	0	dense_98[0][0]
n_hi_ts_block_7 (NHiTSBlock)	[(None, 1320), (None, 64)]	761,784	reshape_7[0][0]
subtract_7 (Subtract)	(None, 1320)	0	subtract_6[0][0], n_hi_ts_block_7[0][0]
dense_103 (Dense)	(None, 1320)	1,743,720	subtract_7[0][0]
reshape_8 (Reshape)	(None, 60, 22)	0	dense_103[0][0]
n_hi_ts_block_8 (NHiTSBlock)	[(None, 1320), (None, 64)]	761,784	reshape_8[0][0]
add_6 (Add)	(None, 64)	0	n_hi_ts_block_6[0][0], n_hi_ts_block_7[0][0], n_hi_ts_block_8[0][0]
dense_108 (Dense)	(None, 64)	4,168	add_6[0][0]
dropout_93 (Dropout)	(None, 64)	0	dense_108[0][0]
future_input (InputLayer)	(None, 3)	0	-
concatenate_17 (Concatenate)	(None, 67)	0	dropout_93[0][0], future_input[0][0]
dense_109 (Dense)	(None, 1)	68	concatenate_17[0][0]



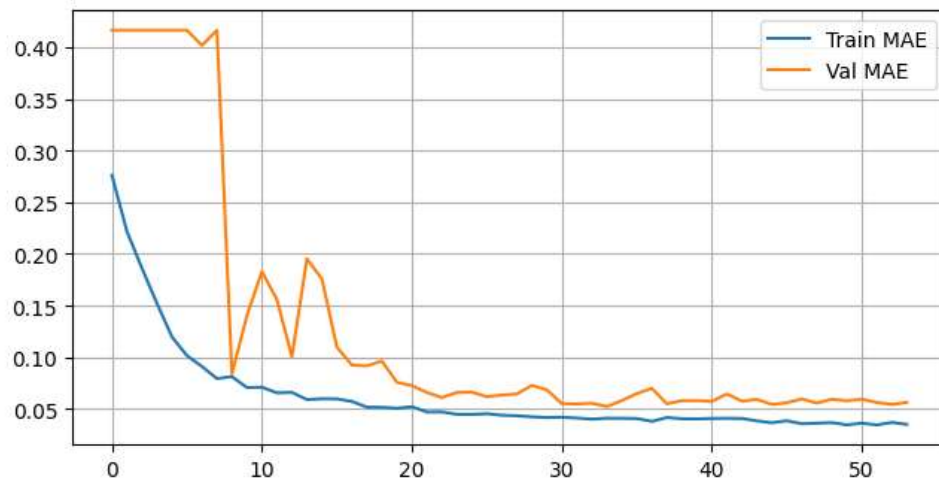
N-HiTS

N-HiTS Training History

N-HiTS — Loss



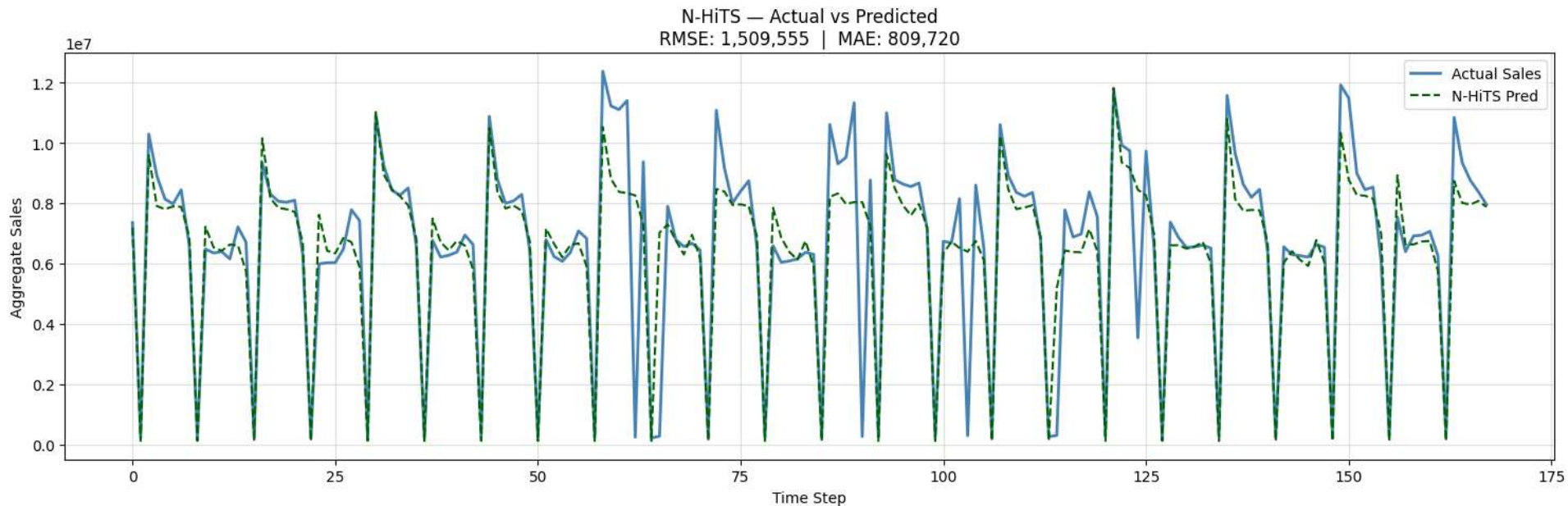
N-HiTS — MAE



Evaluation Metrics : RMSE: 1.51 | MAE: 0.81 | R2: 0.79



N-HiTS Prediction





All Deep Learning Model Comparison

Model	RMSE	MAE	R2
FNN	1.43	0.89	0.81
N - HiTS	1.51	0.81	0.79
LSTM	1.54	0.88	0.78
SimpleRNN	1.54	0.87	0.78
GRU	1.58	0.92	0.76
Patch TST	1.66	1.06	0.74
Bi - LSTM	1.78	1.11	0.70
TCN	1.89	1.09	0.66



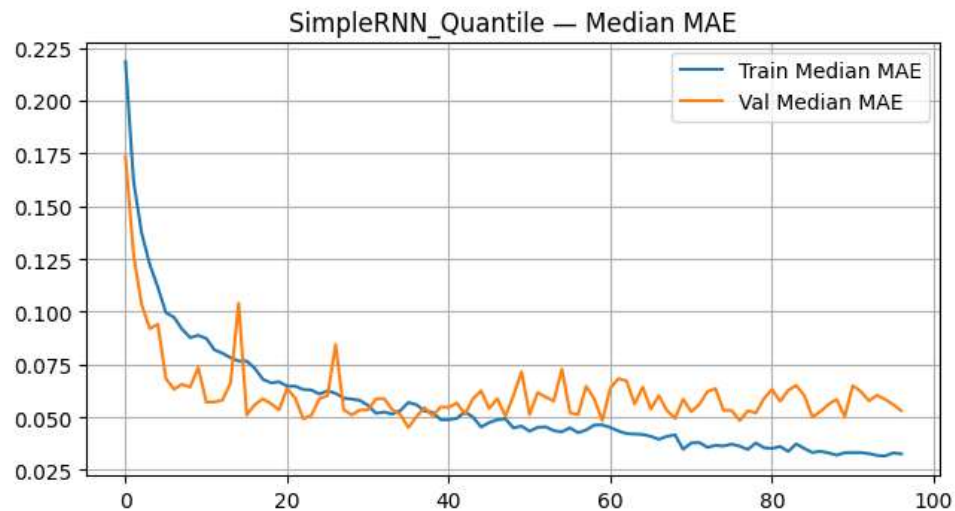
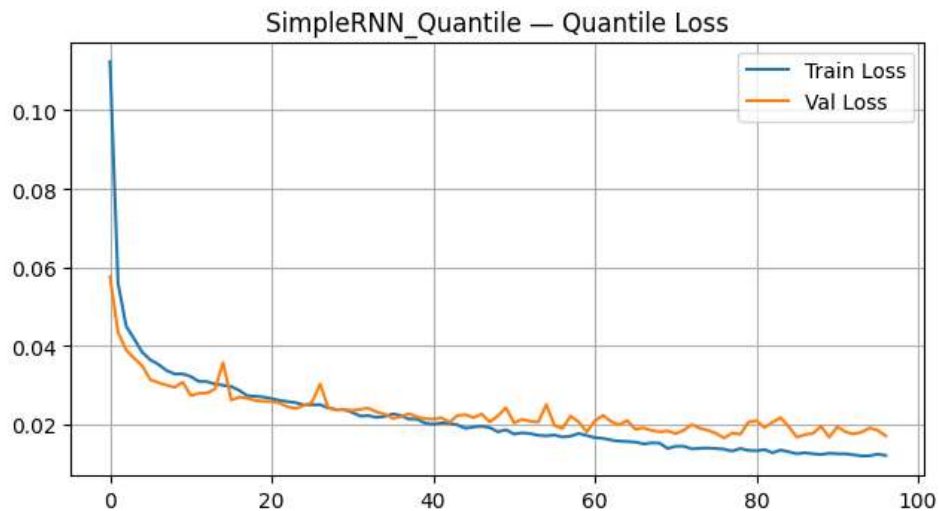
Quantile Regression Models - Quantile RNN

Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 60, 22)	0	-
simple_rnn_6 (SimpleRNN)	(None, 60, 128)	19,328	seq_input[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 60, 128)	512	simple_rnn_6[0][...]
dropout_103 (Dropout)	(None, 60, 128)	0	batch_normalizat...
simple_rnn_7 (SimpleRNN)	(None, 64)	12,352	dropout_103[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 64)	256	simple_rnn_7[0][...]
dropout_104 (Dropout)	(None, 64)	0	batch_normalizat...
dense_121 (Dense)	(None, 32)	2,080	dropout_104[0][0]
future_input (InputLayer)	(None, 3)	0	-
concatenate_19 (Concatenate)	(None, 35)	0	dense_121[0][0], future_input[0][...]
dense_122 (Dense)	(None, 3)	108	concatenate_19[0...



Quantile RNN Loss

SimpleRNN_Quantile Training History

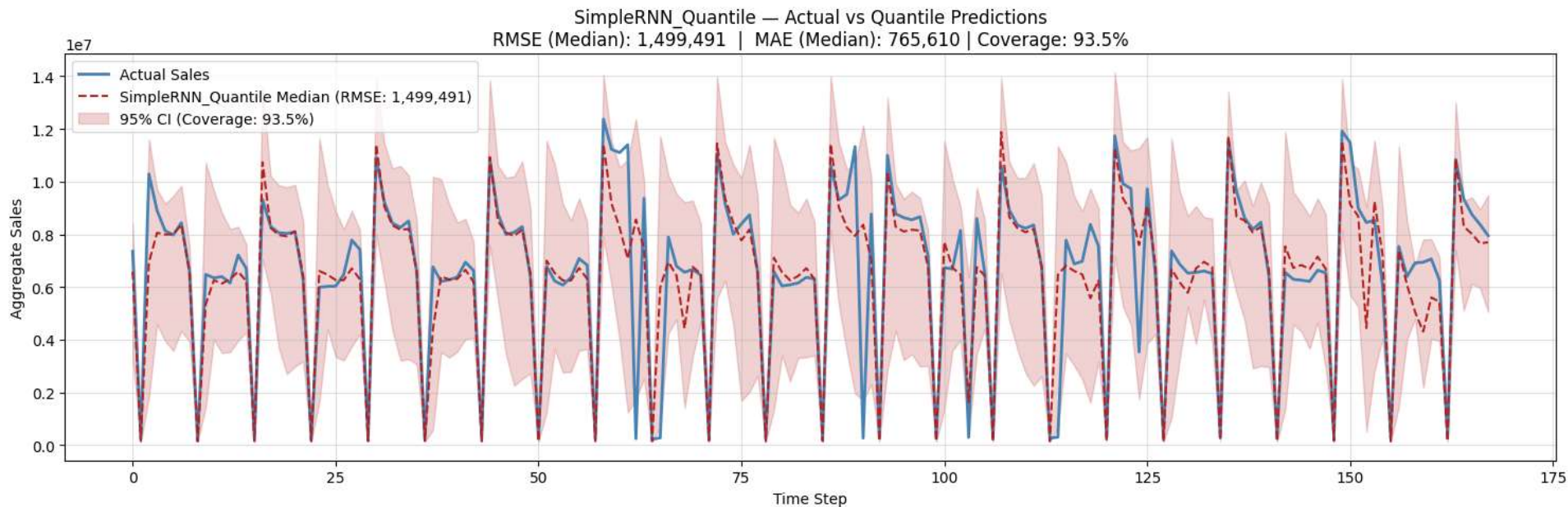




Quantile RNN Prediction

PICP: 93.45%

MPIW: 0.88





Quantile GRU Model Structure

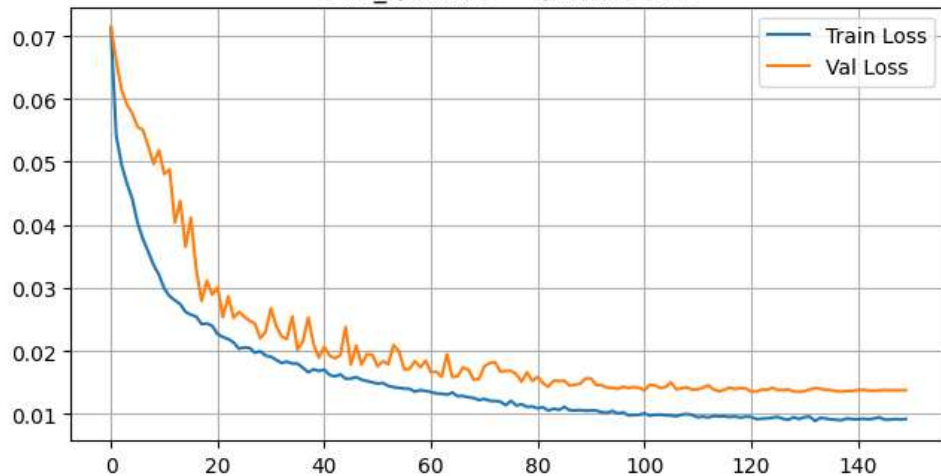
Layer (type)	Output Shape	Param #	Connected to
seq_input (InputLayer)	(None, 60, 22)	0	-
gru_6 (GRU)	(None, 60, 128)	58,368	seq_input[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 60, 128)	512	gru_6[0][0]
dropout_105 (Dropout)	(None, 60, 128)	0	batch_normalizat...
gru_7 (GRU)	(None, 64)	37,248	dropout_105[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 64)	256	gru_7[0][0]
dropout_106 (Dropout)	(None, 64)	0	batch_normalizat...
dense_123 (Dense)	(None, 32)	2,080	dropout_106[0][0]
future_input (InputLayer)	(None, 3)	0	-
concatenate_20 (Concatenate)	(None, 35)	0	dense_123[0][0], future_input[0][...]
dense_124 (Dense)	(None, 3)	108	concatenate_20[0...



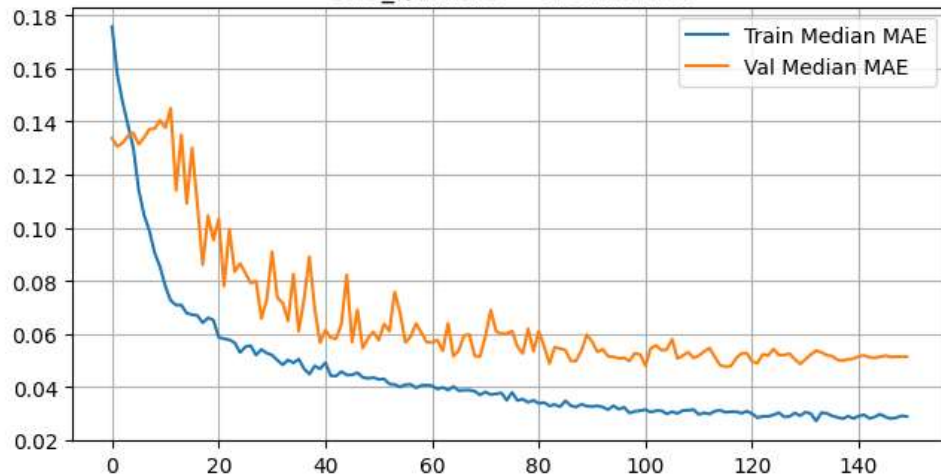
Quantile GRU Loss

GRU_Quantile Training History

GRU_Quantile — Quantile Loss



GRU_Quantile — Median MAE

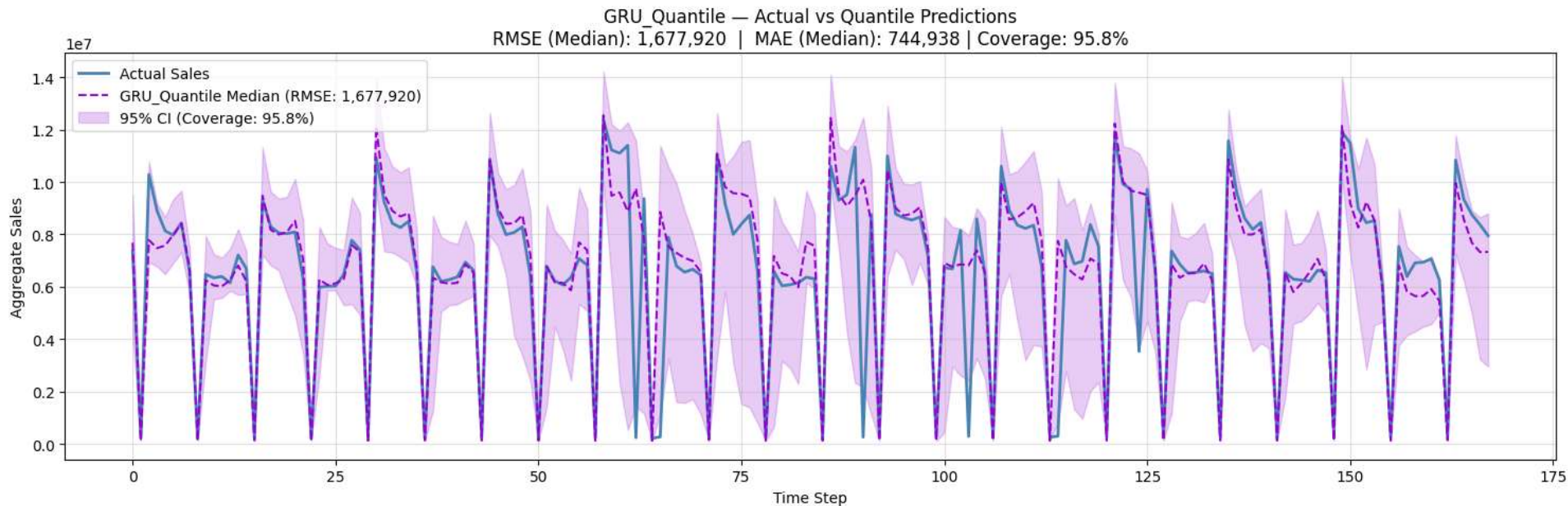




Quantile GRU Prediction

PICP: 95.83%

MPIW: 0.63



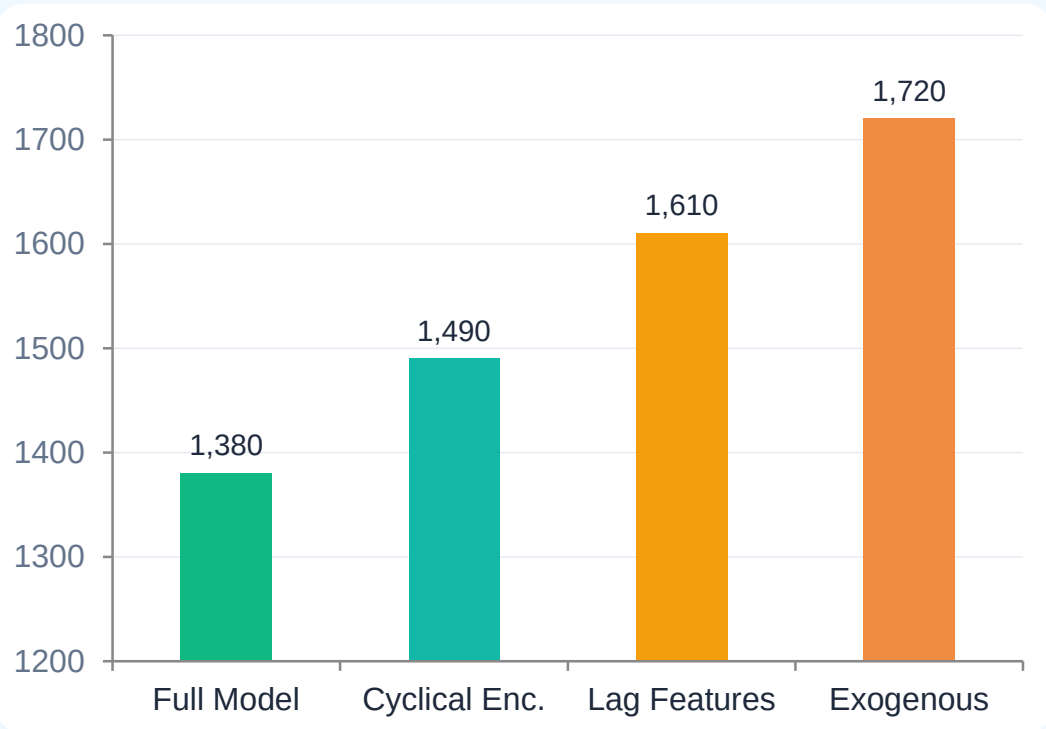


All Quantile Model Comparison

Model	Median RMSE	Median MAE	PICP	MPIW
GRU	1.67	0.74	95.83 %	0.63
RNN	1.50	0.76	93.45 %	0.88
FNN	1.56	0.97	88.69 %	0.95
N - HiTS	1.60	0.82	82.74 %	1.12

Feature Selection

Progressive removal of feature groups from best performing model - impact on RMSE (EUR):



Exogenous Features

+24.6% RMSE

Promo, SchoolHoliday, CompetitionDistance — dominant driver of sales spikes

Lag Features

+16.7% RMSE

Captures short term autocorrelation in the sales signal

Cyclical Encodings

+8.0% RMSE

sin/cos of DayOfWeek & Month outperform plain integer encoding



Github Repository: <https://tinyurl.com/it402Github>



Time Series Models Live Demo: <https://tinyurl.com/it402ts>



Deep Learning Models Live Demo: <https://tinyurl.com/it402-DL-model>

Thank You